Activity 2.5 – Traffic Light Controller

1.Design Approach:

Designed a traffic light controller using Verilog HDL to control north and east road traffic signals. Implemented red, green and yellow light conditions using a finite state machine and timer-based transitions.

2.Verification Steps:

- Created a separate testbench.
- Applied reset to initialize the controller.
- Checked red, green and yellow signal outputs.
- Verified timer-based state transitions.
- Observed the traffic light sequence.
- Simulated using ModelSim.

3.Results:

The testbench checks traffic light sequencing and timing behaviour. PASS/FAIL messages are displayed.

4.Issues Faced:

Timer values must be configured correctly. Traffic signals must follow the correct sequence. Conflicting green signals must be avoided.

5.Conclusion:

Successfully designed and verified a timer-based traffic light controller using Verilog HDL.